BIS 629: Homework 2

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Statement on the use of AI

(a) **How AI was used.** Claude (Anthropic; model identifier `claude-opus-5-5`) was used to check my answer, LaTeX symbols and code.

(b) **Prompts.** See Appendix B.

(c) **Verification.** AI was used to prepare the LaTeX write-up, translate my analysis steps into R code, and review the complete homework. I ran the code in RStudio using the provided dataset and compared the output with the values reported in the write-up, including the cluster sizes, ICC, regression estimates, and power calculations. I also checked that the calculations used the cluster size specified in the question, $m = N/(2J)$. For parts (d)–(f), I entered the relevant parameters into PASS and compared its output with the R results. The required numbers of clusters agreed. For the detectable relative risks, I entered the corresponding probabilities into PASS and confirmed that the displayed powers matched the targets. I have identified the approximation used for unequal cluster sizes and explained the small differences between the R and PASS power calculations.

The supplied data contain $N = 1{,}879$ participants from $J = 14$ clinics. There are no missing values in the analysis variables, and participant IDs are unique. Throughout, $Y = 1$ denotes viral load $\geq 1{,}000$ copies/mL; $X = 1$ denotes early ART. All tests are two-sided with $\alpha = 0.05$.

Group	$Y = 1$	$Y = 0$	Total
Standard care ($X = 0$)	331	582	913
Early ART ($X = 1$)	478	488	966
Total	809	1070	1879

The crude risks are $331/913 = 0.362541$ and $478/966 = 0.494824$.

(a) ICC and cluster sizes

After sorting by clinic, I fit the identity-link Gaussian GEE $E(Y \mid X) = \gamma_0 + \gamma_1 X$ with an exchangeable working correlation. The estimated within-clinic residual correlation is

$$\widehat{\rho} = 0.070114 \quad \text{(robust SE} = 0.023553).$$

The clinic sizes are $109, 130, 167, 121, 62, 110, 132, 463, 104, 137, 70, 79, 149, 46$. Thus,

$$\bar{m} = \frac{1879}{14} = 134.2143, \qquad s_m = \sqrt{\frac{\sum_{j=1}^{14}(m_j - \bar{m})^2}{13}} = 100.7290, \qquad CV_m = \frac{s_m}{\bar{m}} = 0.750509.$$

I use the treatment-adjusted ICC for the subsequent calculations. An intercept-only Gaussian GEE gives 0.075485, a slightly larger marginal ICC. *Verification:* The clinic counts sum to 1,879 and range from 46 to 463; the displayed sample-SD formula agrees with R's `sd()`. The exchangeable logistic GEE gives a similar correlation, 0.070518.

(b) Effective sample size

Using the usual design-effect approximation,

$$D_c = 1 + (\bar{m} - 1)\widehat{\rho}, \qquad D_v = 1 + \{\bar{m}(1 + CV_m^2) - 1\}\widehat{\rho}, \qquad ESS = \frac{N}{D}.$$

Cluster sizes	Design effect	ESS
Constant	10.3402	181.72
Variable	15.6407	120.14

The variable-size approximation retains only 6.39% of the nominal sample size and reduces ESS by 33.9% relative to the constant-size approximation. A future study should recruit more clinics and avoid a few disproportionately large clinics. These ESS values summarize the clustering penalty; they are not a reconstruction of the information in the original stepped-wedge treatment comparison. *Verification:* Direct substitution gives $1 + (134.214286 - 1)(0.070114083) = 10.340198$ and $1879/10.340198 = 181.717999$. Repeating this calculation with D_v gives 120.135412, matching the executed R output.

(c) Intervention effect and Wald tests

I fit $\text{logit}\{P(Y = 1 \mid X)\} = \beta_0 + \beta^* X$ using logistic GEE. The clustered analysis uses clinic IDs and exchangeable correlation; the unclustered analysis uses a separate ID for each participant. Both use sandwich standard errors. The Wald statistic is $W = (\widehat{\beta}^*/SE)^2 \sim \chi_1^2$ under $H_0 : \beta^* = 0$.

Analysis	$\widehat{\beta}^*$	Robust SE	W	p-value	OR (95% CI)
No clustering	0.54365	0.09424	33.281	7.98×10^{-9}	1.722 (1.432, 2.072)
Clinic clustering	0.55244	0.11424	23.385	1.33×10^{-6}	1.737 (1.389, 2.174)

Accounting for clustering increases the SE by 21.2% and widens the CI, but the association remains statistically significant. With the supplied coding, early ART is associated with higher odds of $Y = 1$. This is the unadjusted association requested in the question; the model does not adjust for calendar time. *Verification:* The crude log odds ratio is $\log\{(478 \times 582)/(488 \times 331)\} = 0.5436474$, matching the unclustered GEE and ordinary logistic-regression estimates. The code also recomputes each W, p-value and CI from the coefficient and sandwich covariance matrix.

Planning assumptions for (d)–(f)

Let group 1 be early ART and group 2 be standard care. From the clustered logistic GEE,

$$\widehat{\beta}_0 = -0.454395, \qquad \widehat{\beta}^* = 0.552438,$$
$$p_2 = \text{expit}(\widehat{\beta}_0) = 0.388316, \qquad p_1 = \text{expit}(\widehat{\beta}_0 + \widehat{\beta}^*) = 0.524491.$$

The corresponding pilot OR is 1.737485 and RR is 1.350680.

Following the assignment's stated planning values literally,

$$K_1 = K_2 = k_0 = \frac{J}{2} = 7, \qquad m_1 = m_2 = m = \frac{N}{J \times 2} = 67.107143.$$

Thus the planning design has 14 clusters and expected total enrollment $2k_0 m = 939.5$, rather than 1,879. For variable sizes I retain the pilot $CV_m = 0.750509$, giving a planning SD of 50.3645. This scales the cluster-size distribution to the new mean. I assume equal allocation, a common ICC across arms,

no additional dropout, noninformative cluster size, and a parallel two-arm CRT. All calculations use large-sample normal Wald tests, without continuity or small-sample corrections.

For the planning mean m, the four design effects are

	Constant sizes	Variable sizes
$\rho = 0$	1	1
$\rho = 0.070114$	5.635042	8.285285

(d) Power

For $K_1 = K_2 = k$, the approximate variance of the log odds ratio under the alternative is

$$V(k; p_1, p_2, D) = \frac{D}{km}\left\{\frac{1}{p_1(1-p_1)} + \frac{1}{p_2(1-p_2)}\right\}. \tag{1}$$

Writing $\delta = |\operatorname{logit}(p_1) - \operatorname{logit}(p_2)|/\sqrt{V}$ and $z_{.975} = 1.959964$, the two-sided power is

$$\mathcal{P}(k; p_1, p_2, D) = \Phi(\delta - z_{.975}) + \Phi(-\delta - z_{.975}). \tag{2}$$

At $k = 7$, power is 98.67% without clustering, 42.06% with clustering and constant sizes, and 30.57% with clustering and variable sizes. Variable sizes do not change power when $\rho = 0$ and expected enrollment is held fixed.

Simulation verification: I generated 500 parallel CRTs with seven clinics per arm and 67 participants per clinic (seed 629). Within each arm, clinic probabilities were drawn from a beta distribution with mean p_g and shape sum $(1-\rho)/\rho$, then individual outcomes were drawn conditionally as Bernoulli variables. This independently generates binary outcomes with ICC ρ. Each trial was fitted with exchangeable logistic GEE. Only the simulation rounds m to 67.

Quantity	Result
Simulations successfully fitted	500 / 500
Formula SE at $m = 67$	0.31405
Empirical SD of $\widehat{\beta}^*$	0.31129
Mean sandwich SE	0.28467
Normal-formula power at $m = 67$	0.42049
Empirical GEE rejection rate	0.52600
Monte Carlo SE of rejection rate	0.02233

The empirical SD checks the planning variance under a different data-generating implementation. The empirical rejection rate describes the uncorrected finite-sample GEE Wald test under this alternative; it is not an exact validation of nominal power or type I error. Main design calculations retain the prespecified normal approximation. In this simulation the mean sandwich SE is below the empirical SD, consistent with the observed rejection rate being higher than the planning formula predicts.

(e) Minimum number of clusters

For each target q, I find the smallest integer k satisfying $\mathcal{P}(k; p_1, p_2, D) \geq q$ and report $K = 2k$ total clusters. The mean size, CV, ICC and pilot probabilities remain fixed while k varies. For comparison, dropping only the opposite rejection tail gives the closed-form continuous approximation

$$k_{\text{closed}} = \frac{(z_{.975} + z_q)^2 D}{m(\beta^*)^2} \left\{ \frac{1}{p_1(1-p_1)} + \frac{1}{p_2(1-p_2)} \right\}.$$

The reported integers come from the full two-tail equation, not from assuming that this approximation is an exact inversion.

	80% target		90% target	
Design	k per arm	Achieved power	k per arm	Achieved power
No clustering (either size assumption)	4	88.43%	5	94.18%
Clustering, constant sizes	18	80.54%	24	90.28%
Clustering, variable sizes	27	81.31%	35	90.05%

Verification: Closed-form and numerical continuous solutions differ by less than 10^{-4} clusters per arm and give the same integer requirements. The code prints the achieved power at k and $k-1$; reducing k by one fails to reach the target in every case.

For $\rho = 0$, these are cluster-equivalent counts at the specified mean enrollment; independent individuals, rather than clinics, determine the information.

(f) Minimum detectable RR

I hold $p_2 = 0.388316$, $k = 7$ and $m = 67.107143$ fixed and set $p_1 = RR\, p_2$. For each target q, I solve

$$\mathcal{P}(7; RR\, p_2, p_2, D) = q.$$

The main results use $RR > 1$, the direction observed in the supplied data. "Minimum detectable" means the solution closest to 1 in that direction. The test remains a Wald test of the log odds ratio; RR is used to express the detectable alternative, rather than changing the test to a log-RR test.

Both upward and protective alternatives are reported in the table in (g). For example, the variable-size clustered design requires an RR of at least 1.6854 in the observed direction, or a reduction to approximately 0.3809, to attain 80% power. *Verification:* Substitution of all eight upward and all eight downward RR solutions returns their targets to within 10^{-7}. Searching outward from $RR = 1$ avoids choosing a second root near a probability boundary. The pilot model-based RR is 1.350680; the independently computed crude RR is 1.364877.

(g) Comparison of (d)–(f)

			Total clusters K		Detectable RR (> 1 / < 1)	
Clustering	Size assumption	Power	80%	90%	80%	90%
Ignored	Constant	98.67%	8	10	1.2337 / 0.7758	1.2710 / 0.7416 [1]
Ignored	Variable	98.67%	8	10	1.2337 / 0.7758	1.2710 / 0.7416
Considered	Constant	42.06%	36	48	1.5634 / 0.4852	1.6535 / 0.4083
Considered	Variable	30.57%	54	70	1.6854 / 0.3809	1.7950 / 0.2839

(h) Interpretation and comparison with MaxART

Ignoring clustering overstates power (98.67% versus 30.57%) and understates the cluster requirements. With variable sizes, the planning design needs 54 or 70 total clinics for 80% or 90% power. These counts are about 50% and 46% larger than under constant sizes. The planned 14-clinic parallel design is therefore underpowered for both targets.

Using the original mean size 134.2143 and 14 clinics gives 45.10% power for constant sizes and 32.07% for variable sizes, so the original enrollment would also be insufficient for this hypothetical parallel design. Under the simplified parallel-CRT assumptions used in this homework, the 14-clinic design would be under-powered rather than over-powered. However, the actual MaxART study used a stepped-wedge design, so this comparison should be interpreted only as an approximation rather than a formal prospective power assessment of the original study. The significant result in (c) does not itself establish adequate prospective power.

[1] Background parameters: two-sided $\alpha = 0.05$; equal allocation; $m = 67.107143$ per clinic; $\rho = 0.070114$ when clustering is considered and zero otherwise; $CV_m = 0$ for constant sizes and 0.750509 for variable sizes; no additional dropout; normal log-odds-ratio Wald test. For power and detectable RR, $K_1 = K_2 = 7$. For power and cluster counts, $p_1 = 0.524491$ and $p_2 = 0.388316$ ($\beta_0 = -0.454395$, $\beta^* = 0.552438$). For detectable RR, p_2 is fixed and p_1 varies. Reported K is total across both arms and is rounded to an even integer.

Appendix A. Check against the documented PASS method

Verification status. The main calculations were performed in R 4.3.3 with geepack 1.3.10. I then entered the corresponding parameters into PASS to independently check the power, required number of clusters, and detectable RR. The PASS screenshots are included at the end of the appendix. Small differences in power are due to the approximation used by PASS.

The relevant procedure is *GEE Tests for Multiple Proportions in a Cluster-Randomized Design*, Chapter 534 of the PASS manual [**?**]. With two groups, contrast $(1, -1)$, no missing data and equal allocation, its contrast variance reduces to equation (1) for constant sizes. For the variable-size design-effect approximation, an equivalent constant-size input is

$$\rho_{\text{eq}} = \frac{D_v - 1}{m - 1} = 0.1102042,$$

which reproduces the assumed variance. This is a variance-matching device, not a re-estimate of the biological ICC or a validation of the unequal-size approximation.

The manual's sample-size inversion uses the dominant-tail normal approximation, corresponding to $\mathcal{P}_{\text{PASS}} \approx \Phi(\delta - z_{.975})$. The main calculations retain both rejection tails as in equation (2).

Design	Equivalent ICC	Two-tail power	PASS-formula power
No clustering (either size assumption)	0	0.986665	0.986665
Clustering, constant sizes	0.0701141	0.420583	0.420483
Clustering, variable sizes	0.1102042	0.305676	0.305352

The two implementations give identical integer cluster requirements $(8, 10)$, $(36, 48)$ and $(54, 70)$ at the 80% and 90% targets. Detectable RRs also agree to four decimal places. As an external numerical check, the code reproduces Chapter 534, Example 1: $G = 3$, $k_g = 10$, $m = 10$, $(p_g) = (0.4, 0.2, 0.2)$, contrast $(-2, 1, 1)$, $\rho = 0.6$ and $\alpha = 0.05$ give power 0.3000777, which rounds to the published 0.3001.

Inputs for direct PASS checking. Select Chapter 534, two groups, equal cluster allocation, $m = 67.107143$, proportions $(0.5244911761, 0.3883162647)$, contrast $(1, -1)$, $\alpha = 0.05$, and missing proportion 0. Use ICC 0, 0.0701140833 or 0.1102042015 for the three distinct scenarios above. For (d), solve for power with $k_g = 7$; for (e), solve for total clusters with target power 0.80 or 0.90. For (f), keep $k_g = 7$ and search over $p_1 = RR \times 0.3883162647$ until the target is reached, selecting the crossing closest to $RR = 1$ in the requested direction. If an installed version requires integer mean sizes, use its unrounded formula for comparison or report the rounding change; 67 is not exactly 67.107143.

Appendix B. AI prompts

1. Preparing the LaTeX document. Please use the assignment PDF and the solution draft I provided to prepare a LaTeX document organized by question. Review the draft's proposed methods, retain the reasonable ideas, and flag any steps that need correction or clarification. Present the formulas, calculations, and conclusions clearly, and follow the submission requirements shown in the screenshots.

2. Generating and checking the R code. Please translate the analysis steps in the draft into runnable R code using the provided dataset. Compare the output screenshots I provide with the results in the LaTeX document and update any discrepancies.

3. Reviewing the complete homework. Please check the complete write-up against every part of the assignment and identify any missing answers, unsupported assumptions, or incorrect reasoning. Check that the formulas, R code, software output, and written conclusions are consistent. Explain any issues you find so I can review the proposed changes.

Appendix C. Complete executed R code

```
if (!requireNamespace("geepack", quietly = TRUE)) {
  stop('Install geepack first: install.packages("geepack")')
}
args <- commandArgs(trailingOnly = TRUE)
data_file <- if (length(args)) args[1] else "MaxArt_HW2_2026.RData"
if (!file.exists(data_file) && interactive()) data_file <- file.choose()
env <- new.env()
loaded <- load(data_file, envir = env)
candidates <- loaded[vapply(loaded, function(nm) {
  x <- get(nm, env)
  is.data.frame(x) && all(c("Y", "treatment", "clinc", "id") %in% names(x))
}, logical(1))]
stopifnot(length(candidates) == 1L)
d <- get(candidates, env)
stopifnot(!anyNA(d[c("Y", "treatment", "clinc", "id")]),
          !anyDuplicated(d$id), all(d$Y %in% 0:1),
          all(d$treatment %in% 0:1))
d <- d[order(d$clinc, d$id), ]
d$clinic_id <- as.integer(factor(d$clinc))
d$person_id <- seq_len(nrow(d))
outdir <- "results"
dir.create(outdir, showWarnings = FALSE)
N <- nrow(d)
sizes <- as.numeric(table(d$clinc))
J <- length(sizes)
m_observed <- mean(sizes)
sd_observed <- sd(sizes)
cv <- sd_observed / m_observed
# (a) Gaussian identity-link exchangeable GEE; alpha estimates residual ICC.
fit_icc <- geepack::geeglm(Y ~ treatment, data = d, id = clinic_id,
  family = gaussian("identity"), corstr = "exchangeable")
fit_icc_marginal <- geepack::geeglm(Y ~ 1, data = d, id = clinic_id,
  family = gaussian("identity"), corstr = "exchangeable")
rho <- unname(fit_icc$geese$alpha)
rho_marginal <- unname(fit_icc_marginal$geese$alpha)
rho_se <- unname(sqrt(fit_icc$geese$valpha[1,1]))
# (b) Conventional design-effect approximation for individual-weighted means.
DE <- function(m, rho, cv = 0) 1 + (m * (1 + cv^2) - 1) * rho
de_observed <- c(constant = DE(m_observed, rho),
                 variable = DE(m_observed, rho, cv))
ess <- N / de_observed
# (c) The no-clustering analysis assigns a distinct cluster to every person.
# An independence correlation is equivalent to exchangeable for singletons
# and avoids attempting to estimate a nonidentifiable singleton ICC.
fit_no <- geepack::geeglm(Y ~ treatment, data = d, id = person_id,
  family = binomial("logit"), corstr = "independence")
fit_yes <- geepack::geeglm(Y ~ treatment, data = d, id = clinic_id,
  family = binomial("logit"), corstr = "exchangeable")
stopifnot(fit_icc$geese$error == 0, fit_yes$geese$error == 0,
          fit_no$geese$error == 0)
extract <- function(fit, label) {
  b <- unname(coef(fit)["treatment"])
  se <- unname(sqrt(diag(vcov(fit)))[2])
  z <- b / se
  data.frame(analysis = label, beta0 = unname(coef(fit)[1]),
    beta = b, SE = se, Z = z, Wald = z^2,
    p_value = 2 * pnorm(-abs(z)), OR = exp(b),
    CI_low = exp(b - qnorm(.975) * se),
    CI_high = exp(b + qnorm(.975) * se))
}
model_results <- rbind(extract(fit_no, "No clustering"),
                       extract(fit_yes, "Clinic clustering"))
beta0 <- unname(coef(fit_yes)[1])
beta <- unname(coef(fit_yes)[2])
p2 <- plogis(beta0)                 # standard of care, X=0
p1 <- plogis(beta0 + beta)          # early ART, X=1
# (d)-(g) Follow the assignment literally: m = N / J / 2, K1=K2=J/2.
# For unequal sizes retain the observed CV (rather than the observed SD),
# so the entire size distribution scales with the new mean.
alpha <- .05
k0 <- J / 2
m <- N / J / 2
zcrit <- qnorm(1 - alpha / 2)
```

```
designs <- data.frame(
  scenario = c("No clustering: constant", "No clustering: variable",
               "Clustering: constant", "Clustering: variable"),
  rho = c(0, 0, rho, rho), cv = c(0, cv, 0, cv))
designs$DE <- mapply(DE, m = m, rho = designs$rho, cv = designs$cv)
variance_logOR <- function(k, m, p1, p2, D) {
  D / (k * m) * (1 / (p1 * (1-p1)) + 1 / (p2 * (1-p2)))
}
wald_power <- function(k, m, p1, p2, D, both_tails = TRUE) {
  delta <- abs(qlogis(p1) - qlogis(p2)) /
    sqrt(variance_logOR(k, m, p1, p2, D))
  main <- pnorm(delta - zcrit)
  if (both_tails) main + pnorm(-delta - zcrit) else main
}
minimum_k <- function(target, m, p1, p2, D, both_tails = TRUE) {
  f <- function(k) wald_power(k, m, p1, p2, D, both_tails) - target
  hi <- 2
  while (f(hi) < 0) hi <- 2 * hi
  continuous <- uniroot(f, c(1e-8, hi), tol = 1e-10)$root
  k <- max(1L, ceiling(continuous))
  stopifnot(f(k) >= -1e-10, k == 1L || f(k-1L) < 0)
  c(continuous = continuous, per_arm = k, total = 2*k,
    achieved_power = wald_power(k, m, p1, p2, D, both_tails))
}
# Find the detectable RR closest to 1. Scanning is necessary because
# log-odds Wald power is not globally monotone near p1=0 or p1=1.
detectable_RR <- function(target, k, m, p2, D, direction = "increase",
                          both_tails = TRUE) {
  end <- if (direction == "increase") 1 - 1e-8 else 1e-8
  grid <- seq(p2, end, length.out = 2001)
  f <- function(p) wald_power(k, m, p, p2, D, both_tails) - target
  v <- vapply(grid, f, numeric(1))
  hit <- which(v >= 0)[1]
  if (is.na(hit)) return(NA_real_)
  root <- uniroot(f, sort(grid[c(hit-1, hit)]), tol = 1e-12)$root
  stopifnot(abs(f(root)) < 1e-7)
  root / p2
}
designs$power <- vapply(designs$DE, function(D)
  wald_power(k0, m, p1, p2, D), numeric(1))
for (target in c(.8, .9)) {
  tag <- as.character(100 * target)
  ks <- t(vapply(designs$DE, function(D)
    minimum_k(target, m, p1, p2, D), numeric(4)))
  designs[[paste0("k", tag)]] <- ks[, "per_arm"]
  designs[[paste0("K", tag)]] <- ks[, "total"]
  designs[[paste0("achieved", tag)]] <- ks[, "achieved_power"]
  designs[[paste0("RR", tag)]] <- vapply(designs$DE, function(D)
    detectable_RR(target, k0, m, p2, D, "increase"), numeric(1))
  designs[[paste0("RRprotective", tag)]] <- vapply(designs$DE, function(D)
    detectable_RR(target, k0, m, p2, D, "decrease"), numeric(1))
}
# Verification: GLM point estimate, analytic sample-size inversion,
# achieved power at k and k-1, and substitution of both RR roots.
glm_check <- glm(Y ~ treatment, data=d, family=binomial("logit"))
glm_difference <- max(abs(coef(glm_check)-coef(fit_no)))
stopifnot(glm_difference < 1e-8)
checks <- do.call(rbind, lapply(c(.8,.9), function(target) {
  do.call(rbind,lapply(seq_len(nrow(designs)),function(i) {
    D <- designs$DE[i]
    ans <- minimum_k(target,m,p1,p2,D)
    closed <- (zcrit+qnorm(target))^2*D/(m*beta^2)*
      (1/(p1*(1-p1))+1/(p2*(1-p2)))
    k <- ans["per_arm"]
    up <- detectable_RR(target,k0,m,p2,D,"increase")
    dn <- detectable_RR(target,k0,m,p2,D,"decrease")
    data.frame(scenario=designs$scenario[i],target=target,
      k_closed=closed,k_numeric=unname(ans["continuous"]),
      k_integer=unname(k),power_at_k=wald_power(k,m,p1,p2,D),
      power_at_k_minus1=wald_power(k-1,m,p1,p2,D),
      power_at_RR_up=wald_power(k0,m,up*p2,p2,D),
      power_at_RR_down=wald_power(k0,m,dn*p2,p2,D))
  }))
}))
stopifnot(max(abs(checks$k_closed-checks$k_numeric)) < 1e-4,
  all(checks$power_at_k >= checks$target),
```

```
  all(checks$power_at_k_minus1 < checks$target),
  max(abs(checks$power_at_RR_up-checks$target)) < 1e-7,
  max(abs(checks$power_at_RR_down-checks$target)) < 1e-7)
# Independent Monte Carlo check of the constant-size planning variance.
# Beta-binomial generation gives Corr(Y_ij,Y_ij') = rho within a clinic.
# m is rounded to 67 for this simulation only; main calculations retain N/J/2.
set.seed(629)
B <- 500L
mm <- round(m)
shape <- (1-rho)/rho
sim_results <- t(replicate(B, {
  pp <- c(rbeta(k0,p2*shape,(1-p2)*shape),
          rbeta(k0,p1*shape,(1-p1)*shape))
  ss <- data.frame(cl=rep(seq_len(2*k0),each=mm),
    x=rep(rep(0:1,each=k0),each=mm))
  ss$y <- rbinom(nrow(ss),1,pp[ss$cl])
  ff <- try(suppressWarnings(geepack::geeglm(y~x,id=cl,data=ss,
    family=binomial("logit"),corstr="exchangeable")),silent=TRUE)
  if (inherits(ff,"try-error") || ff$geese$error != 0) {
    c(beta=NA_real_,SE=NA_real_,reject=NA_real_)
  } else {
    b <- unname(coef(ff)[2]); se <- sqrt(vcov(ff)[2,2])
    if (!all(is.finite(c(b,se))) || se<=0)
      c(beta=NA_real_,SE=NA_real_,reject=NA_real_)
    else c(beta=b,SE=se,reject=as.numeric(abs(b/se)>zcrit))
  }
}))
good <- complete.cases(sim_results)
stopifnot(sum(good)>0.95*B)
sim_results <- sim_results[good,,drop=FALSE]
rejection_rate <- mean(sim_results[,"reject"])
mc_check <- c(B_requested=B,B_used=nrow(sim_results),mean_size=mm,
  formula_SE=sqrt(variance_logOR(k0,mm,p1,p2,DE(mm,rho))),
  empirical_SD=sd(sim_results[,"beta"]),
  mean_robust_SE=mean(sim_results[,"SE"]),
  formula_power=wald_power(k0,mm,p1,p2,DE(mm,rho)),
  empirical_rejection=rejection_rate,
  rejection_MCSE=sqrt(rejection_rate*(1-rejection_rate)/nrow(sim_results)))
# Independent implementation of PASS Chapter 534's contrast-variance formula.
# PASS's documented sample-size inversion uses the dominant-tail approximation.
pass_power <- function(probs, contrast, clusters, m, rho) {
  total <- sum(clusters)
  allocation <- clusters / total
  h <- (1 + (m-1)*rho) / m
  V <- diag(h / (allocation * probs * (1-probs)))
  delta <- abs(sum(contrast * qlogis(probs))) /
    sqrt(drop(t(contrast) %*% V %*% contrast) / total)
  pnorm(delta - qnorm(1-alpha/2))
}
# Official PASS manual, Example 1, first row: reported power 0.3001.
published_check <- pass_power(c(.4,.2,.2), c(-2,1,1), c(10,10,10), 10, .6)
stopifnot(round(published_check, 4) == .3001)
pass_checks <- designs[c("scenario", "DE")]
pass_checks$rho_equivalent <- (designs$DE - 1) / (m - 1)
pass_checks$PASS_formula_power <- vapply(pass_checks$rho_equivalent,
  function(rr) pass_power(c(p1,p2), c(1,-1), c(k0,k0), m, rr), numeric(1))
pass_checks$two_tail_power <- designs$power
for (target in c(.8,.9)) {
  tag <- as.character(100 * target)
  pass_checks[[paste0("PASS_formula_K",tag)]] <- vapply(designs$DE,
    function(D) minimum_k(target,m,p1,p2,D,FALSE)["total"], numeric(1))
  pass_checks[[paste0("PASS_formula_RR",tag)]] <- vapply(designs$DE,
    function(D) detectable_RR(target,k0,m,p2,D,"increase",FALSE), numeric(1))
}
# Sensitivity: use all observed participants in a parallel two-arm design.
full_size <- designs[c("scenario", "rho", "cv")]
full_size$m <- m_observed
full_size$DE <- mapply(DE, full_size$m, full_size$rho, full_size$cv)
full_size$power <- vapply(full_size$DE, function(D)
  wald_power(k0, m_observed, p1, p2, D), numeric(1))
cluster_summary <- aggregate(cbind(Y,treatment) ~ clinc, d, mean)
cluster_summary$n <- as.integer(table(d$clinc)[as.character(cluster_summary$clinc)])
write.csv(cluster_summary, file.path(outdir,"clinic_summary.csv"),row.names=FALSE)
write.csv(model_results,file.path(outdir,"gee_results.csv"),row.names=FALSE)
write.csv(designs,file.path(outdir,"design_results.csv"),row.names=FALSE)
write.csv(pass_checks,file.path(outdir,"pass_formula_checks.csv"),row.names=FALSE)
```

```
write.csv(full_size,file.path(outdir,"full_size_sensitivity.csv"),row.names=FALSE)
write.csv(checks,file.path(outdir,"verification_checks.csv"),row.names=FALSE)
write.csv(data.frame(parameter=names(mc_check),value=as.numeric(mc_check)),
  file.path(outdir,"monte_carlo_check.csv"),row.names=FALSE)
parameters <- c(N=N,J=J,mean_cluster=m_observed,sd_cluster=sd_observed,CV=cv,
  ICC=rho,ICC_SE=rho_se,ICC_intercept_only=rho_marginal,
  ICC_logistic=fit_yes$geese$alpha,
  DE_observed_constant=de_observed[1],DE_observed_variable=de_observed[2],
  ESS_constant=ess[1],ESS_variable=ess[2],
  beta0=beta0,beta=beta,p1=p1,p2=p2,RR=p1/p2,
  future_k_per_arm=k0,future_m=m,future_SD=cv*m,alpha=alpha,
  PASS_published_example=published_check)
write.csv(data.frame(parameter=names(parameters),value=as.numeric(parameters)),
  file.path(outdir,"parameters.csv"),row.names=FALSE)
sink(file.path(outdir,"analysis_output.txt"), split=TRUE)
options(digits=7,width=95,useFancyQuotes=FALSE)
cat("INPUTS AND DATA VALIDATION\n");print(parameters)
cat("\nObserved 2 x 2 table (rows: treatment; columns: Y)\n")
print(addmargins(table(d$treatment,d$Y)))
cat("No missing analysis variables; participant IDs unique.\n")
cat("\n(a) Linear GEE ICC fit\n");print(summary(fit_icc))
options(digits=7)
cat("\n(b) Observed design effects and ESS\n")
print(data.frame(DE=de_observed,ESS=ess))
cat("\n(c) Logistic GEE: coefficients and robust Wald tests\n")
print(model_results[c("analysis","beta0","beta","SE","Wald","p_value")],row.names=FALSE)
print(model_results[c("analysis","OR","CI_low","CI_high")],row.names=FALSE)
cat("Maximum no-cluster GEE vs GLM coefficient difference:",glm_difference,"\n")
cat("\n(d) Planning design effects, SE and power\n")
planning <- designs[c("scenario","DE","power")]
planning$SE <- sqrt(vapply(designs$DE,function(D)
  variance_logOR(k0,m,p1,p2,D),numeric(1)))
print(planning,row.names=FALSE)
cat("\n(e) Minimum cluster counts and achieved power\n")
print(designs[c("scenario","k80","K80","achieved80","k90","K90","achieved90")],row.names=FALSE)
cat("\n(f) Detectable RR closest to 1 in each direction\n")
print(designs[c("scenario","RR80","RR90","RRprotective80","RRprotective90")],row.names=FALSE)
cat("\n(g) Summary\n")
print(designs[c("scenario","power","K80","K90","RR80","RR90")],row.names=FALSE)
cat("\nVERIFICATION: closed-form versus two-tail numerical inversion\n")
print(checks[c("scenario","target","k_closed","k_numeric")],row.names=FALSE)
cat("\nVERIFICATION: integer minima and substitution of both RR roots\n")
print(checks[c("scenario","target","power_at_k","power_at_k_minus1",
  "power_at_RR_up","power_at_RR_down")],row.names=FALSE)
cat("\nVERIFICATION: beta-binomial Monte Carlo (seed=629)\n");print(mc_check)
cat("\nPASS DOCUMENTED-FORMULA CHECKS (not PASS software output)\n")
print(pass_checks[c("scenario","rho_equivalent","PASS_formula_power","two_tail_power")],row.names=FALSE)
print(pass_checks[c("scenario","PASS_formula_K80","PASS_formula_K90")],row.names=FALSE)
print(pass_checks[c("scenario","PASS_formula_RR80","PASS_formula_RR90")],row.names=FALSE)
cat("\n(h) Sensitivity with observed mean cluster size\n")
print(full_size[c("scenario","m","DE","power")],row.names=FALSE)
cat("\nSESSION INFORMATION\n");print(sessionInfo())
sink()
saveRDS(list(icc=fit_icc,icc_intercept_only=fit_icc_marginal,
  no_clustering=fit_no,clustering=fit_yes),file.path(outdir,"fitted_models.rds"))
```

Appendix D. Actual R output

```
INPUTS AND DATA VALIDATION
                            N                             J                  mean_cluster
                1879.00000000                   14.00000000                   134.21428571
                   sd_cluster                            CV                            ICC
                  100.72901848                    0.75050892                    0.07011408
                       ICC_SE            ICC_intercept_only            ICC_logistic.alpha
                    0.02355344                    0.07548457                    0.07051808
DE_observed_constant.constant DE_observed_variable.variable         ESS_constant.constant
                   10.34019752                   15.64068386                   181.71799874
        ESS_variable.variable                         beta0                          beta
                  120.13541200                   -0.45439525                    0.55243842
                            p1                            p2                            RR
                    0.52449118                    0.38831626                    1.35068042
              future_k_per_arm                     future_m                     future_SD
                   7.00000000                   67.10714286                   50.36450924
                         alpha        PASS_published_example
                    0.05000000                    0.30007773

Observed 2 x 2 table (rows: treatment; columns: Y)

         0    1  Sum
  0    582  331  913
  1    488  478  966
  Sum 1070  809 1879
No missing analysis variables; participant IDs unique.

(a) Linear GEE ICC fit

Call:
geepack::geeglm(formula = Y ~ treatment, family = gaussian("identity"),
    data = d, id = clinic_id, corstr = "exchangeable")

 Coefficients:
            Estimate Std.err   Wald Pr(>|W|)
(Intercept)  0.38889 0.02892 180.81  < 2e-16 ***
treatment    0.13494 0.02717  24.66 6.85e-07 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate  Std.err
(Intercept)   0.2416 0.005581
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha  0.07011 0.02355
Number of clusters:   14  Maximum cluster size: 463

(b) Observed design effects and ESS
              DE      ESS
constant 10.34020 181.7180
variable 15.64068 120.1354

(c) Logistic GEE: coefficients and robust Wald tests
          analysis      beta0      beta         SE     Wald      p_value
     No clustering -0.5643521 0.5436474 0.09423689 33.28072 7.976954e-09
 Clinic clustering -0.4543953 0.5524384 0.11423947 23.38489 1.326158e-06
          analysis       OR   CI_low  CI_high
     No clustering 1.722277 1.431820 2.071656
 Clinic clustering 1.737485 1.388929 2.173511
Maximum no-cluster GEE vs GLM coefficient difference: 3.630429e-14

(d) Planning design effects, SE and power
                scenario       DE     power        SE
 No clustering: constant 1.000000 0.9866652 0.1322799
 No clustering: variable 1.000000 0.9866652 0.1322799
```

```
    Clustering: constant 5.635042 0.4205830 0.3140092
    Clustering: variable 8.285285 0.3056761 0.3807567

(e) Minimum cluster counts and achieved power
                scenario k80 K80 achieved80 k90 K90 achieved90
 No clustering: constant   4   8  0.8843488   5  10  0.9417506
 No clustering: variable   4   8  0.8843488   5  10  0.9417506
    Clustering: constant  18  36  0.8054375  24  48  0.9027942
    Clustering: variable  27  54  0.8131441  35  70  0.9004883

(f) Detectable RR closest to 1 in each direction
                scenario     RR80     RR90 RRprotective80 RRprotective90
 No clustering: constant 1.233652 1.270974      0.7757966      0.7415997
 No clustering: variable 1.233652 1.270974      0.7757966      0.7415997
    Clustering: constant 1.563361 1.653504      0.4851646      0.4082940
    Clustering: variable 1.685414 1.794987      0.3809334      0.2838898

(g) Summary
                scenario     power K80 K90     RR80     RR90
 No clustering: constant 0.9866652   8  10 1.233652 1.270974
 No clustering: variable 0.9866652   8  10 1.233652 1.270974
    Clustering: constant 0.4205830  36  48 1.563361 1.653504
    Clustering: variable 0.3056761  54  70 1.685414 1.794987

VERIFICATION: closed-form versus two-tail numerical inversion
                scenario target  k_closed k_numeric
 No clustering: constant    0.8  3.150110  3.150102
 No clustering: variable    0.8  3.150110  3.150102
    Clustering: constant    0.8 17.751000 17.750956
    Clustering: variable    0.8 26.099557 26.099493
 No clustering: constant    0.9  4.217103  4.217102
 No clustering: variable    0.9  4.217103  4.217102
    Clustering: constant    0.9 23.763553 23.763544
    Clustering: variable    0.9 34.939902 34.939890

VERIFICATION: integer minima and substitution of both RR roots
                scenario target power_at_k power_at_k_minus1 power_at_RR_up power_at_RR_down
 No clustering: constant    0.8  0.8843488         0.7805524            0.8              0.8
 No clustering: variable    0.8  0.8843488         0.7805524            0.8              0.8
    Clustering: constant    0.8  0.8054375         0.7828108            0.8              0.8
    Clustering: variable    0.8  0.8131441         0.7985003            0.8              0.8
 No clustering: constant    0.9  0.9417506         0.8843488            0.9              0.9
 No clustering: variable    0.9  0.9417506         0.8843488            0.9              0.9
    Clustering: constant    0.9  0.9027942         0.8904735            0.9              0.9
    Clustering: variable    0.9  0.9004883         0.8920781            0.9              0.9

VERIFICATION: beta-binomial Monte Carlo (seed=629)
       B_requested             B_used          mean_size          formula_SE
       500.00000000       500.00000000         67.00000000          0.31405067
       empirical_SD      mean_robust_SE       formula_power empirical_rejection
         0.31128500          0.28467297          0.42049236          0.52600000
     rejection_MCSE
         0.02233043

PASS CHECKS
                scenario rho_equivalent PASS_formula_power two_tail_power
 No clustering: constant     0.00000000          0.9866652      0.9866652
 No clustering: variable     0.00000000          0.9866652      0.9866652
    Clustering: constant     0.07011408          0.4204831      0.4205830
    Clustering: variable     0.11020420          0.3053523      0.3056761
                scenario PASS_formula_K80 PASS_formula_K90
 No clustering: constant                8               10
 No clustering: variable                8               10
    Clustering: constant               36               48
    Clustering: variable               54               70
                scenario PASS_formula_RR80 PASS_formula_RR90
 No clustering: constant          1.233652          1.270974
 No clustering: variable          1.233652          1.270974
    Clustering: constant          1.563361          1.653505
    Clustering: variable          1.685415          1.794987

(h) Sensitivity with observed mean cluster size
```

```
               scenario        m       DE     power
 No clustering: constant 134.2143  1.00000 0.9999603
 No clustering: variable 134.2143  1.00000 0.9999603
    Clustering: constant 134.2143 10.34020 0.4510263
    Clustering: variable 134.2143 15.64068 0.3206840

SESSION INFORMATION
R version 4.3.3 (2024-02-29)
Platform: x86_64-pc-linux-gnu (64-bit)
Running under: Ubuntu 24.04.3 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

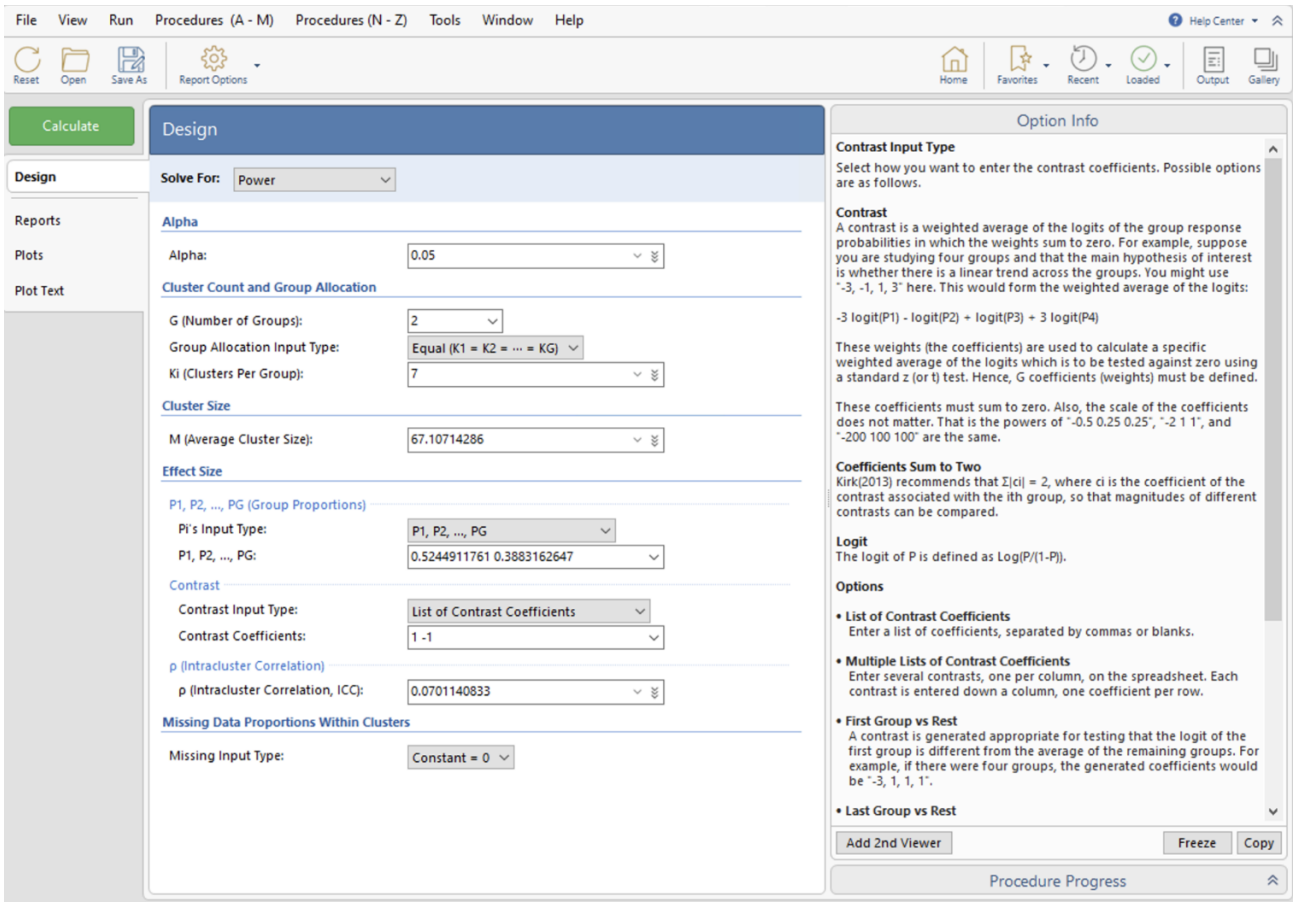
locale:
 [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
 [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
 [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
[10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C

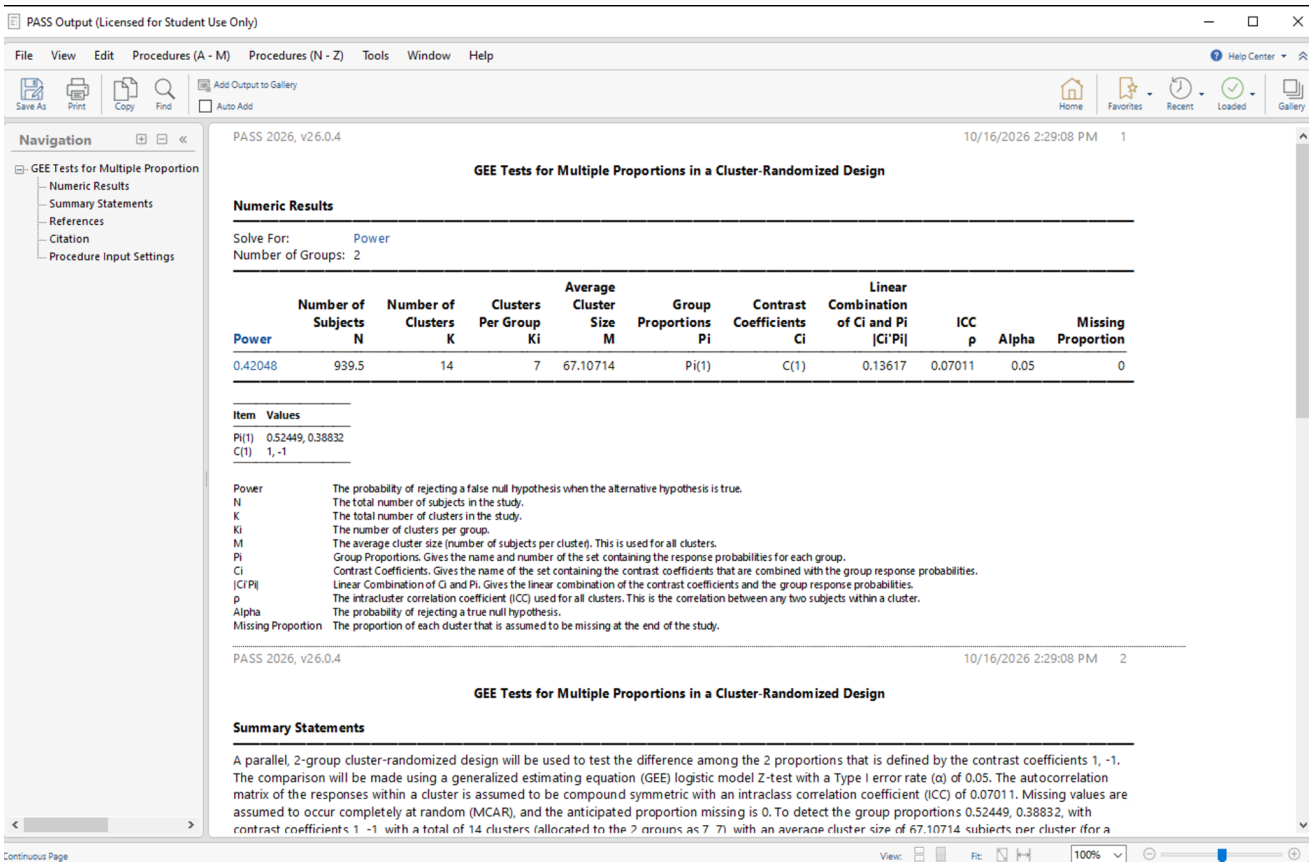
time zone: Pacific/Tarawa
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

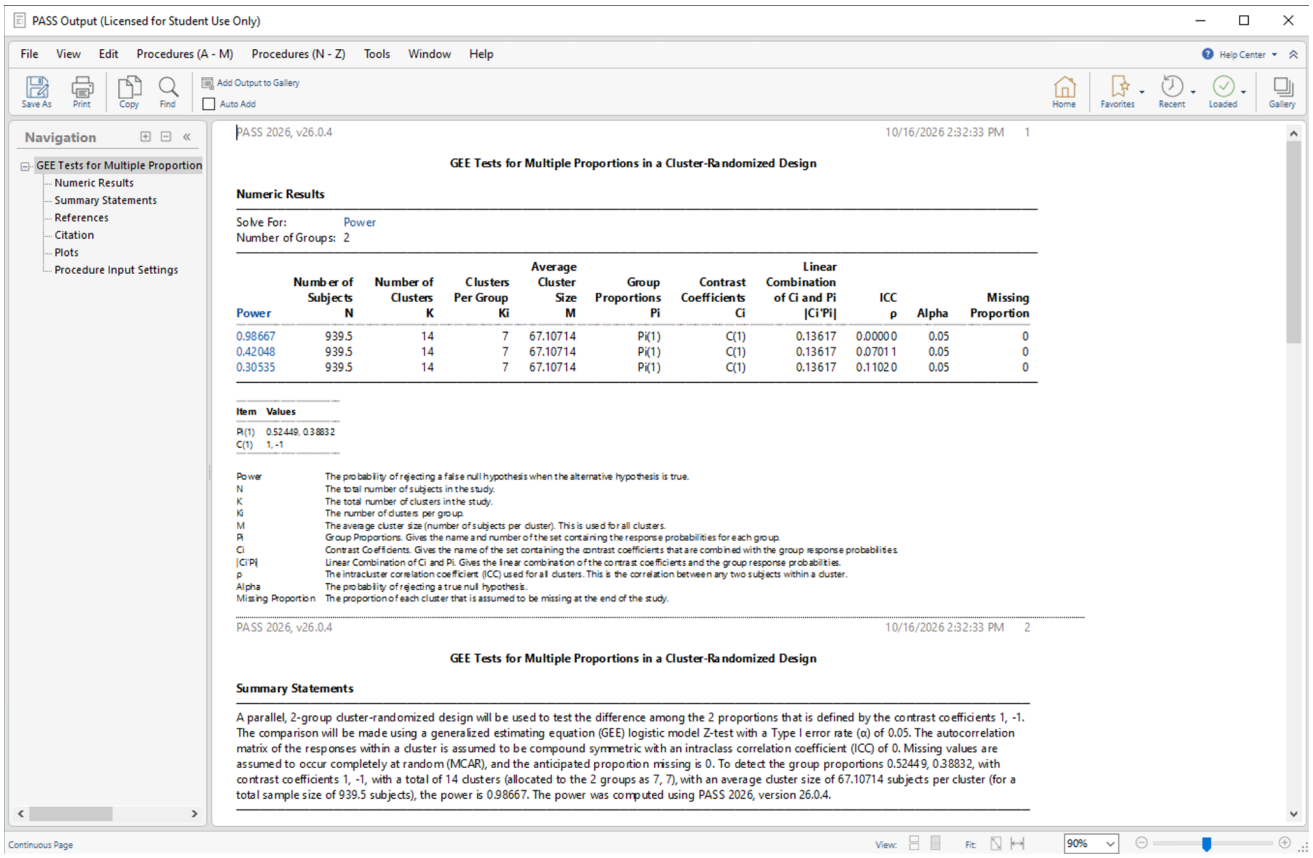
loaded via a namespace (and not attached):
 [1] backports_1.4.1  tidyr_1.3.1      utf8_1.2.4       R6_2.5.1         tidyselect_1.2.0
 [6] magrittr_2.0.3   glue_1.7.0       tibble_3.2.1     pkgconfig_2.0.3  dplyr_1.1.4
[11] generics_0.1.3   lifecycle_1.0.4  cli_3.6.2        fansi_1.0.5      vctrs_0.6.5
[16] compiler_4.3.3   purrr_1.0.2      broom_1.0.5      pillar_1.9.0     geepack_1.3.10
[21] rlang_1.1.3      MASS_7.3-60.0.1
```

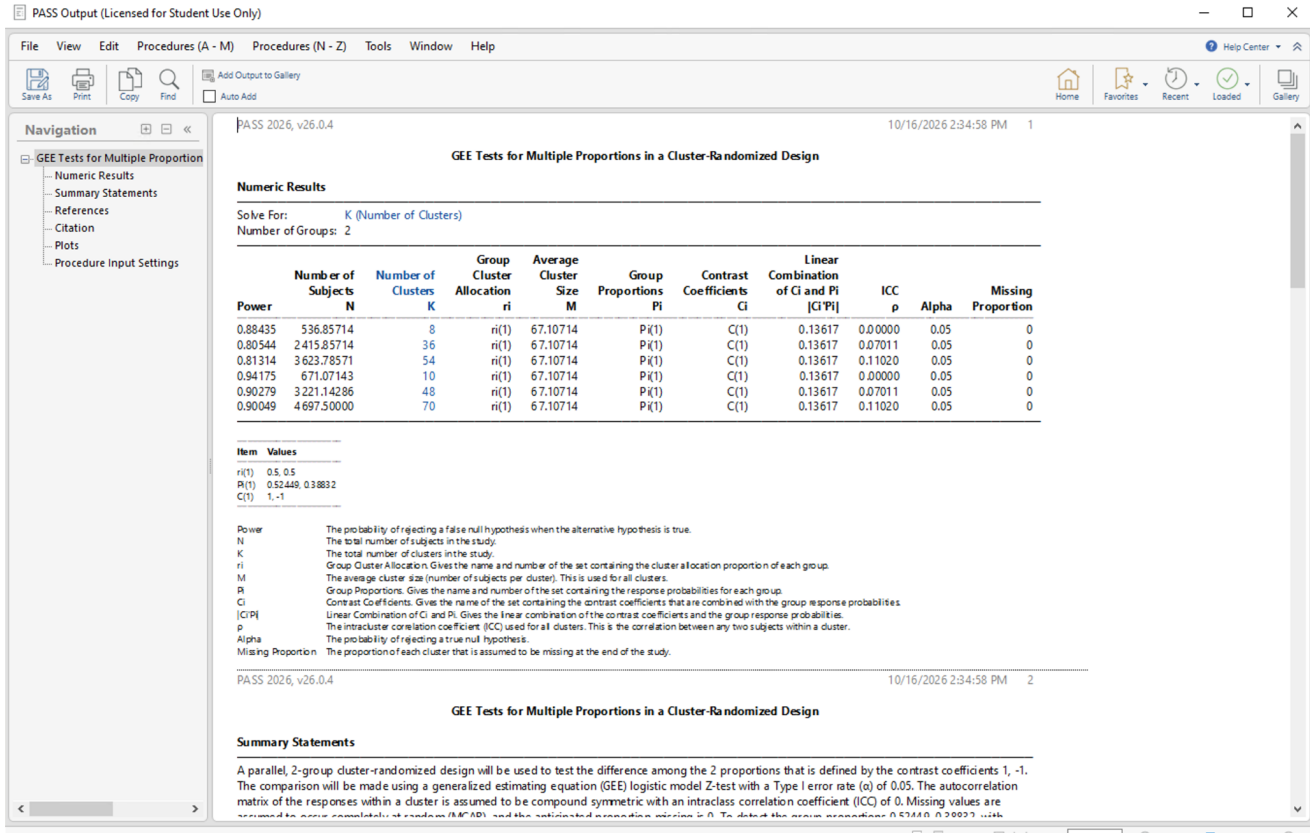
PASS screenshot





PASS screenshot





PASS screenshot

